Sara Robin Wilson

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Education

Michigan State University — East Lansing, MI
Doctor of Philosophy in Computational Mathematics, Science, and Engineering — Expected May 2031
Advised by Prof. Andrew Christlieb and Prof. Brian O'Shea

University of Pittsburgh — Pittsburgh, PA
Bachelor of Science in Mathematics — May 2026

Research Interests

computational plasma physics; numerical analysis of PDEs; scientific machine learning

Honors & Awards

Edna M. Heck Scholarship
Society of Women Engineers Certificate of Merit in Math and Science

Publications

An Analysis of a 2×2 Keyfitz-Kranzer Type Balance System with Varying Generalized Chaplygin Gas — Sept 2024
Physics of Fluids
J. Frew, N. Keyser, E. Kim, G. Paddock, C. Toumbleston, **S. Wilson**, C. Tsikkou
10.1063/5.0231413

Research Experience

Michigan State University Doctoral Research — East Lansing, MI
Research on computational plasma physics. — Jun 2026 - Current

Semiconductor Research Corporation REU — Tuscaloosa, AL
Research on design modifications for wireless power transfer. — Jun 2025 – Jul 2025

West Virginia University REU — Morgantown, WV
Research on numerical analysis of 2×2 Keyfitz-Kranzer type balance system with varying Chaplygin gas. — Jun 2024 – Jul 2024

Skills

Programming & Scientific Computing: Python; MATLAB; Java

Computational Methods: finite element methods; finite differences; scientific machine learning

Tools & Software: Git; LaTeX; ParaView

Talks & Presentations

Directed Reading Program, Carnegie Mellon University — Pittsburgh, PA
Computational Plasma Physics — Dec 2025

Summer Research Symposium, University of Alabama — Tuscaloosa, AL
Wireless Power Transfer — Jul 2025

Pi Mu Epsilon Poster Session, Joint Mathematics Meetings — Seattle, WA
Numerical Analysis of the Riemann Problem for a Cosmological 2×2 Balance System — Jan 2025

Summer Research Symposium, West Virginia University — Morgantown, WV
Numerical Analysis of the Riemann Problem for a Cosmological 2×2 Balance System — Jul 2024

Additional Training

Directed Reading Program, Carnegie Mellon University — Pittsburgh, PA
Computational Plasmas Physics & Nodal Discontinuous Galerkin Methods — Sept 2025 - Apr 2026

Summer School, Princeton University — Princeton, NJ
Fluids and Computer Assisted Proofs — Aug 2025

Summer School, Texas A&M University — College Station, TX
Modeling and Simulation of Partial Differential Equations — May 2025

Teaching Experience

University of Pittsburgh — Pittsburgh, PA
Teaching Assistant — Aug 2024 – Apr 2026

University of Pittsburgh — Pittsburgh, PA
Mathematics Tutor — Jan 2023 – Apr 2026

West Virginia University — Remote
REU Mentor — Jun 2025 - Jul 2025